

Climate Change Doesn't Start with Disasters.

It starts with anomalies.

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Introduction

A temperature anomaly shows how much warmer or colder a place is compared to what is considered normal. Instead of using absolute temperatures, anomalies measure the difference from a

long-term average. In this project, temperatures are compared to the average conditions from 1961 to 1990.

Temperature anomalies are important because they help us understand climate change. Weather can change from day to day, but anomalies reveal long-term trends. They make it possible to compare different regions, even if their climates are very different.

The maps show selected European countries and their mean temperature anomalies over specific time periods. These countries were chosen because clear temperature anomalies occurred at the same time as major events such as heat waves, droughts, forest fires, or water shortages. These events did not happen by chance. Persistent temperature anomalies increase the likelihood and intensity of extreme events.

Together, they show how long-term temperature changes can lead to real impacts that affect people, nature, and infrastructure.

European countries and their climate change related Events

Portugal



Portugal

Between 1991 and 1995, Portugal experienced an unusually hot and dry period. These temperature anomalies were among the earliest signs of the heat waves and droughts that would later affect large parts of the Iberian Peninsula. Reduced rainfall and persistent heat put strong pressure on water resources, agriculture, and daily life.

In some southern regions, water supply interruptions lasted more than 12 hours. The growing scarcity of water highlighted how sensitive Portugal is to long-term temperature anomalies.

This period eventually led to political responses, including the Albufeira Convention between Portugal and Spain, aimed at regulating shared water resources under increasing climatic stress.

Question:

What was an early consequence of temperature anomalies in Portugal in the 1990s?

- A) Glacier retreat
- B) Long-lasting water shortages
- C) Increased winter flooding

Answer **B** is correct! The Text says, that Portugal was really hot and dry, which also means less rain fell. When you have less rain and more heat, there's less water available for people to use and that's why they ran out of water.

This is what "long-lasting water shortage means"

Spain



Spain

Between 2004 and 2008, Spain experienced prolonged droughts linked to strong temperature anomalies. Freshwater lakes and reservoirs dropped to critically low levels, revealing cracked ground where water once stood.

Agricultural areas in central and southern Spain turned from green to brown as soil moisture disappeared. Water restrictions became part of everyday life, even in major cities. Spain's example shows

how sustained temperature anomalies can reshape landscapes and expose a society's dependence on stable water availability.

Question:

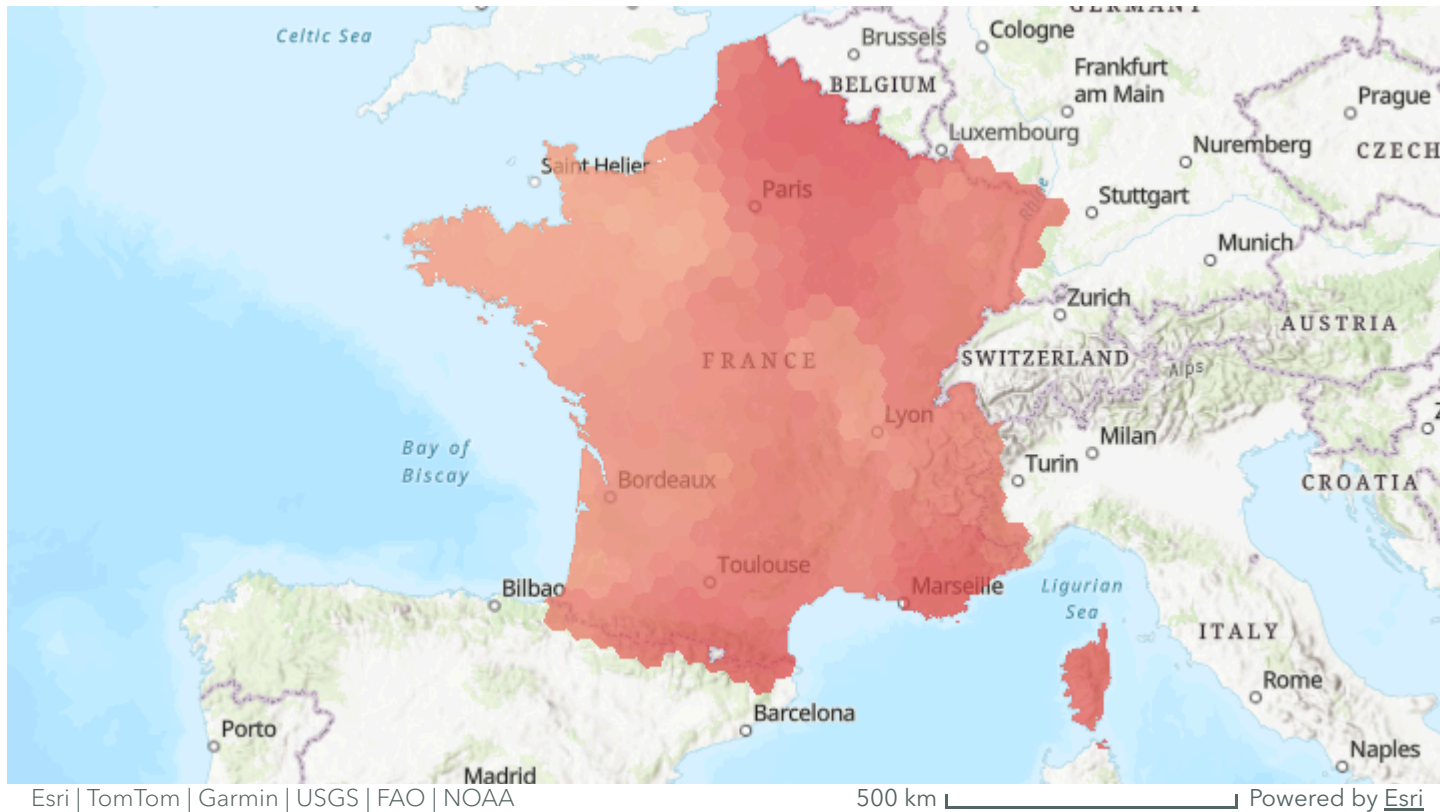
What visible sign made Spain's droughts especially striking during this period?

- A) Flooded coastlines
- B) Cracked lake beds and empty reservoirs
- C) Expanding glaciers

Answer **B** is correct! If a serious drought happens, the water in lakes and reservoirs dries up. It can be so dramatic, that you actually can see the ground underneath crack open.

This is a physical sign that shows how bad the water shortage was.

France



France

Between 2001 and 2004, France experienced a strong increase in temperature, culminating in the extreme heat wave of 2003.

This event affected cities, agriculture, transport systems, and public health across the country.

Urban areas, particularly in southern France, recorded very high night-time temperatures due to dense infrastructure and limited cooling. The event demonstrated that heat waves were becoming more frequent and more severe.

France's experience highlights how temperature anomalies can translate into widespread societal impacts, especially in densely populated and built-up regions.

Question:

Why was the 2003 heat wave a turning point for France?

- A) It affected only rural areas
- B) It revealed growing vulnerability to extreme heat
- C) It caused no long-term impacts

The correct answer is **B!** The heatwave of 2003 showed that the country wasn't prepared for such heat - cities got dangerously hot, especially at night, because of the buildings trapping heat. This affected everything from health to agriculture. It showed, that heatwaves were becoming a real threat and societies need to deal with them.

Italy



Italy

From 2003 to 2006, Italy experienced a series of exceptionally hot summers. The 2003 heat wave severely affected the Po Valley, where riverbeds became visible after decades and water availability dropped sharply.

In cities such as Milan, Turin, and Trento, dense buildings and paved surfaces created urban heat islands. Heat remained trapped well into the night, increasing stress for residents.

Italy's case

demonstrates how temperature anomalies can intensify extreme heat even in regions accustomed to warm climates, especially when urban structures amplify their effects.

Question:

Which effect of temperature anomalies was especially visible in Italian cities?

- A) Heavy snowfall
- B) Urban heat islands
- C) Increased rainfall

Again answer **B** is correct! Urbane areas like the city of Milan have lots of buildings and pavements. This traps heat an makes it even hotter than it would normally be. Especially at night the heat couldn't escape the cities, so there was not so much of cooling down as per usual.

This effect is called "urban heat island" and its a visible effect of how cities can make the temperature anomalies even worse.

Poland



Poland

In the early 1990s, Poland underwent rapid economic growth after the end of communism. This transformation relied heavily on coal as the main energy source for industry, electricity, and household heating.

As a result, CO₂ emissions remained high during this period. Between 1992 and 1994, these conditions coincided with the first measurable warming signals in

Poland. The country illustrates how economic structures and energy choices can influence temperature anomalies.

Poland's case highlights the link between long-term industrial activity and gradual warming trends, rather than single extreme weather events.

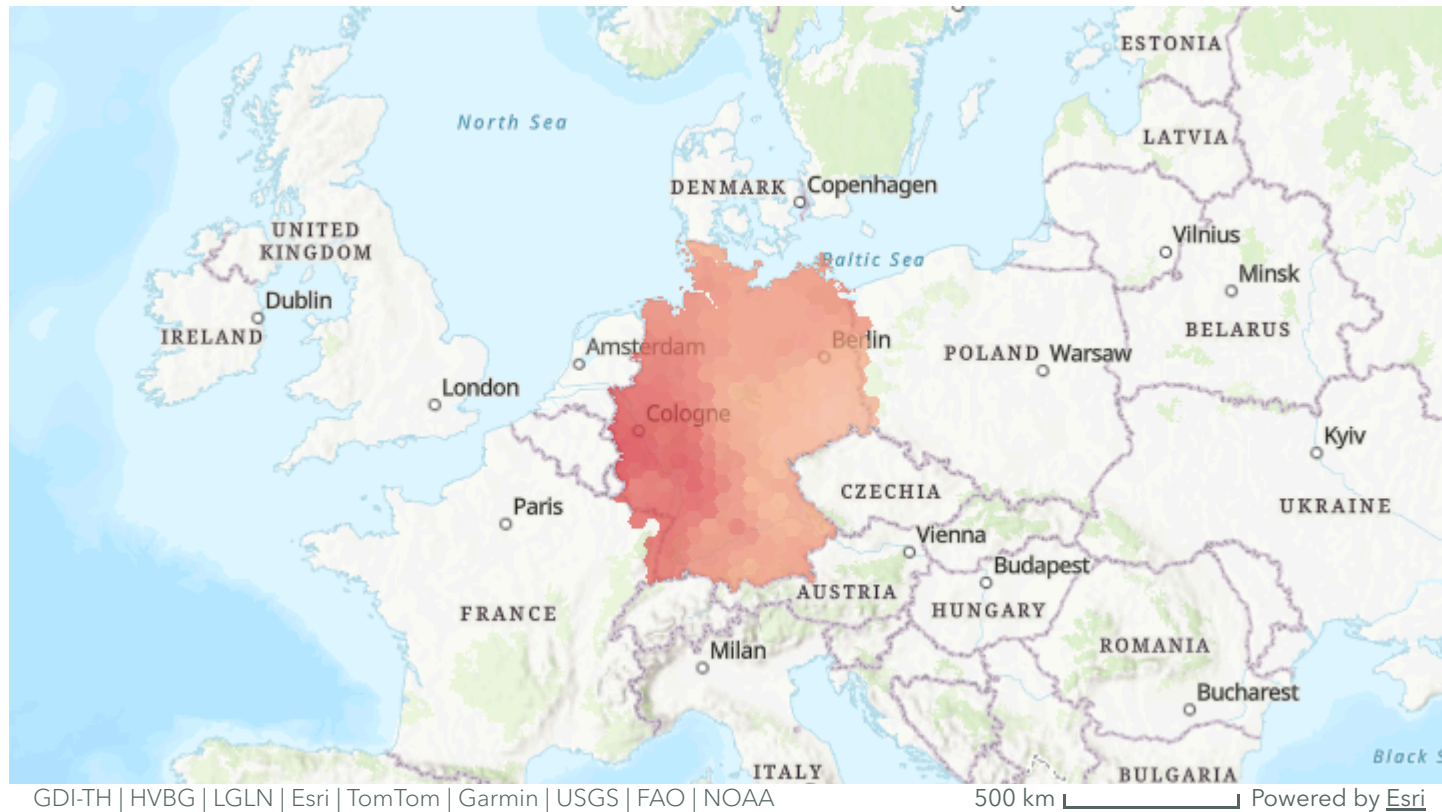
Question:

Why do temperature anomalies appear in Poland during the early 1990s?

- A) Increased use of renewable energy
- B) Strong dependence on coal-based industry
- C) Reduced energy consumption

Answer **B** is correct! In the text it says, that Poland burned a lot of coal for energy and industry after communism ended. As you should know, burning coal releases CO₂ - a greenhouse gas that traps heat in the atmosphere and causes global warming. So therefore the more coal you burn, the more CO₂ you release, the hotter it gets!

Germany



Germany

From 2001 to 2004, Germany experienced rising temperatures, strongly influenced by dense infrastructure, industrial activity, and high energy consumption. The 2003 heat wave exposed the vulnerability of urban and industrial regions to extreme heat.

Cities retained heat during the night, while industrial areas struggled to cool down. Electricity demand increased, further reinforcing heat stress. This period marked a turning point in understanding how built environments can intensify temperature anomalies.

Germany's example shows that climate impacts are not only shaped by global trends but also by local land use and infrastructure.

Question:

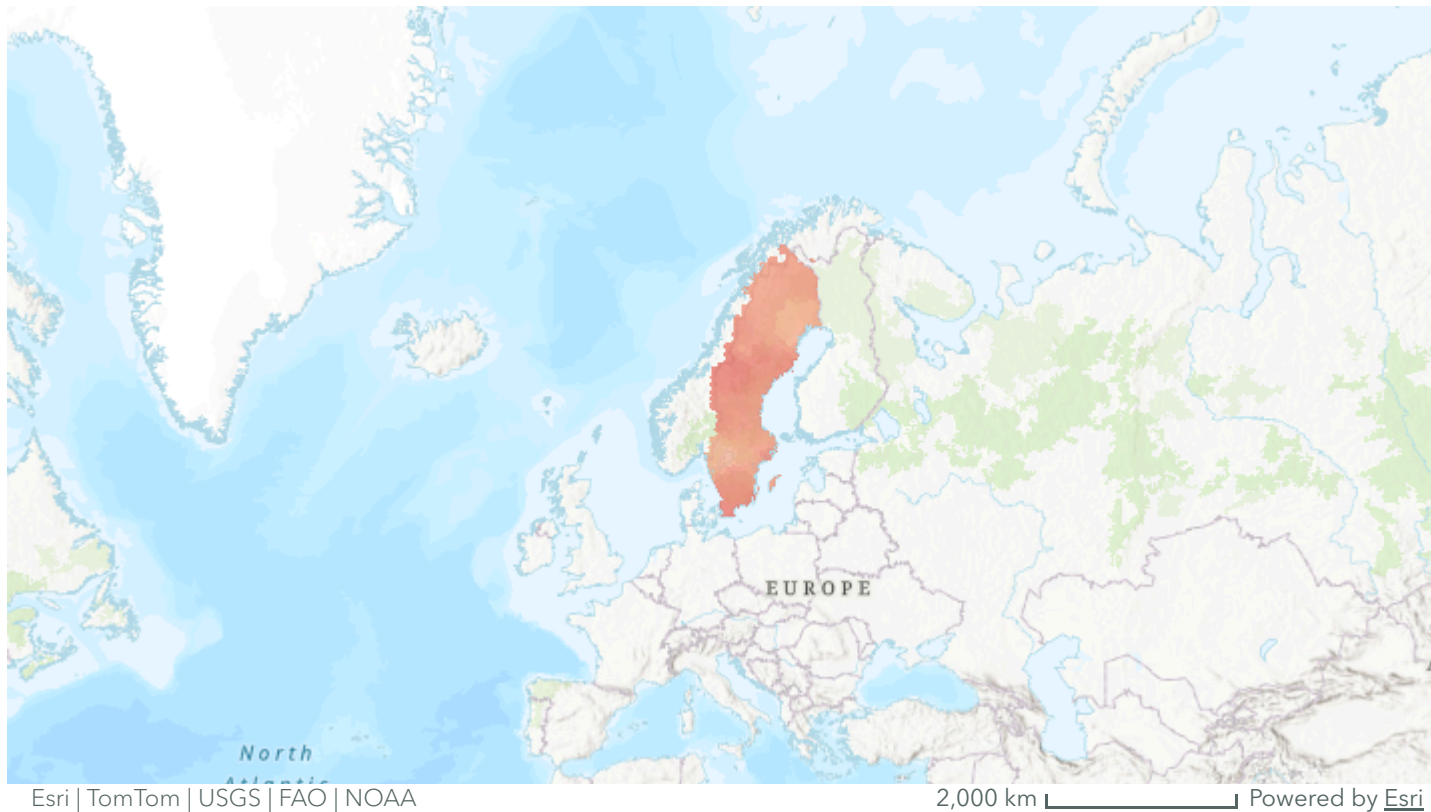
What did the 2003 heat wave reveal about Germany's cities and industry?

- A) They were well adapted to extreme heat
- B) They amplified temperature anomalies
- C) They reduced energy demand

The correct answer is **B**! German cities and factories were so densely built that they would trap heat and couldn't cool down at night. They made the temperature anomalies even worse than it would "naturally" would have been.

The Infrastructure and buildings didn't just experience the heat - they amplified it.

Sweden



Sweden

From 2017 to 2019, Sweden experienced extreme temperature anomalies, with the summer of 2018 breaking multiple heat records. Prolonged drought reduced groundwater levels and caused the worst agricultural harvest since the 1950s.

One striking consequence was the melting of the glacier on Kebnekaise's southern peak, causing it to lose its status as Sweden's highest mountain.

This event became a powerful symbol of how rising temperatures affect even northern, traditionally cold regions.

Question:

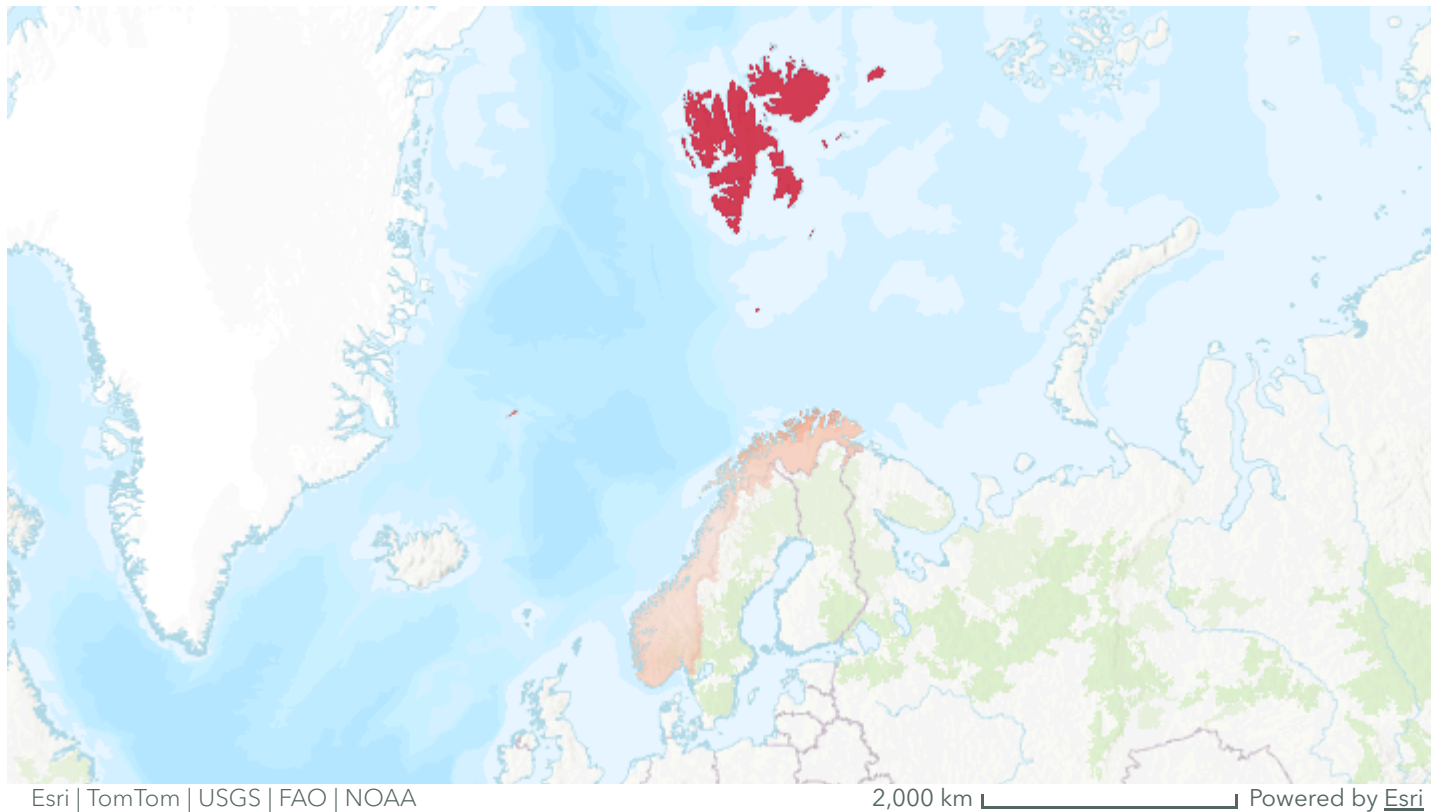
What unprecedented change occurred during Sweden's 2018 heat wave?

- A) A new glacier formed
- B) The country's highest mountain peak changed
- C) Rainfall increased significantly

B is correct because the extreme heat in 2018 was so intense, that it melted the glacier on Sweden's highest mountain. Once the glacier was gone, the peak got shorter and the mountain was no longer the highest in the country.

This is a dramatic and visible change that showed how global warming was affecting even cold places like Sweden.

Norway



Norway

Between 2012 and 2016, Norway confirmed a strong warming trend, with temperatures rising by about 0.5 °C per decade since the late 1970s. These temperature anomalies had visible effects on glaciers, which became smaller than at any time in recent decades.

Earlier snow-melt led to spring floods, while increased evaporation raised the risk of summer droughts.

Norway's case highlights that even cold regions are highly sensitive to long-term warming, with significant impacts on the hydrological cycle.

Question:

How did rising temperatures affect Norway's natural systems?

- A) Growing glaciers
- B) Smaller glaciers and earlier floods
- C) No measurable changes

As always, the correct answer is **B**. The rising temperatures in caused two main effects in Norway: The glaciers melted and got smaller, also the snow melted earlier in the spring than usual. This combined caused flooding.

These are real visible changes that show how warming temperature affect the cold regions.

Sources

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Map Authors	Nikša Borovac; Louis Michael Schmalisch; Markus Tomio; As part of the course work for: 25W856232: Cartographic Design & Geomedia, at the Faculty of Digital and Analytical Sciences of the Paris- Lodron-University of Salzburg. Supervising professors: Priv.-Doz. Dr. Sabine Hennig & D.Sc. Merve Keskin

Basemap dataset	QGIS Development Team. World Map_1.gpkg (GeoPackage). Provided via QGIS; OGRprovider. CRS: WGS 84 / Pseudo-Mercator (EPSG:3857).
Elements	“Save the Earth – Salzburg” stamp created using Artificial Intelligence